

Basics of Multivariate Statistics

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Statistics: everybody uses it, nobody understands it
Why Should you care?
We do Machine Learning not Statistic after all

Definition of Statistics

Statistics (from German: Statistik, orig. "description of a state, a country") is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data

Why is it still important?

- to organize, interpret and understanding data
- to build, select and evaluate models
- contains the building blocks of data science and data analysis
- math at the base of linear regression (neuron)
- powerful (and dangerous) tool to support arguments



- Basics
- Dataset, Mean and Variance
- Least Squared
- Linear Regression
- Bias vs Variance

you're gonna see some math today, but you don't need all of it
these slides have been designed for someone to be able to follow them
without me talking, but also for someone to go back and get a deeper
understanding of what I'm gonna say in person today

Probability Space

$(\Omega, \sigma(\Omega), \mathbb{P})$

- Ω set of all possible single outcomes
- $\sigma(\Omega)$ set of events (all possible combinations of single outcomes)
- $\mathbb{P}$ probability function

Random Variable

A Random Variable $X : \Omega \rightarrow E$ takes as input an outcome and returns a realization

- $(\Omega, \sigma(\Omega), \mathbb{P})$ probability space
- $(E, \mathcal{E})$ measurable space
- $\mathbb{P}(X(\omega) = x)$ is the probability that the realization x of the RV X occurs

Example: Rolling two dice

- $\Omega = \{(1, 1), (1, 2), \dots, (3, 5), \dots, (6, 6)\}$ all possible single events
- $E = \{2, 3, \dots, 12\}$ all possible single realizations
- $X(\omega) = \omega_1 + \omega_2$ is the sum of the two dice
- $\mathbb{P}(X = x) = \begin{cases} 1/36 & \text{for } x = 2, 12 \\ 2/36 & \text{for } x = 3, 11 \\ \vdots & \\ 1/6 & \text{for } x = 7 \end{cases}$



Assume from here both Ω and E are of the form $\mathbb{R}^n$

Probability Density Function

a p.d.f. is a non negative Lebesgue-integrable function

$\mathbb{P}\{a \leq X \leq b\} = \int_a^b f_X(x) dx$ (think of it as the probability in the continuous case)

- $\int_{\mathbb{R}^n} f_X(x) dx = 1$
- $f_X(x) \geq 0 \quad \forall x$



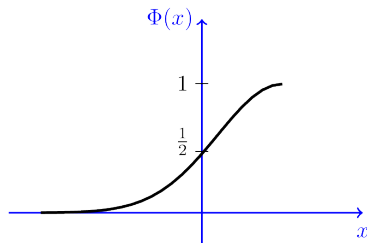
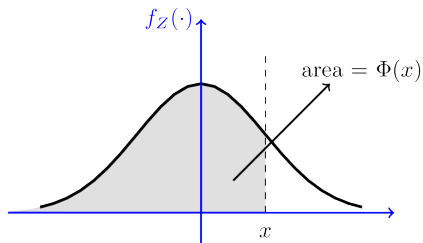
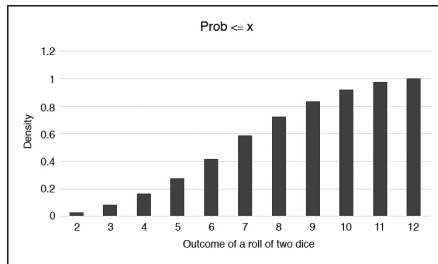
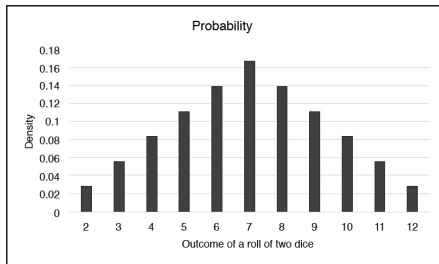
Cumulative Density Function

a c.d.f is $F_X(x) = \int_{-\infty}^x f_X(t) dt = \mathbb{P}(X \leq x)$ (think of it as the sum of the probability of all events up to x)

- $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $\lim_{x \rightarrow +\infty} F_X(x) = 1$
- $f_X(x)$ is right continuous (only look at the limit definition from the right)



Basic Tools for today



Expected Value and Sample Mean

Discrete case:

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x P(X = x)$$

Continuous case:

$$\mathbb{E}[X] = \int_{\mathbb{R}^n} x f(x) dx$$

Sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$



Variance and Sample Variance

Discrete case:

$$\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sum_{x \in \mathcal{X}} (x - \mathbb{E}[X])^2 P(X = x)$$

Continuous case:

$$\text{VAR}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int_{\mathbb{R}^n} (x - \mathbb{E}[X])^2 f(x) dx$$

Sample variance:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Covariance and Sample Covariance

Discrete case:

$$\text{Cov}(X, Y) = \sum_x \sum_y (x - \mathbb{E}[X])(y - \mathbb{E}[Y]) P(X = x, Y = y)$$

Continuous case:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \int_{\mathbb{R}^n} (x - \mathbb{E}[X])(y - \mathbb{E}[Y]) f(x, y) dx$$

Sample Covariance:

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Correlation Coefficient and Sample Correlation

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

$$r_{XY} = \frac{S_{XY}}{S_X S_Y}$$

what does the **Correlation Coefficient** mean?

- $\rho = 0$ means that the two RVs are **independent** and they don't have any influence on each other
- $\rho = 1$ means that the two RVs are completely positively correlated, if one **increases**, the other one **does so one to one**
- $\rho = -1$ means that the two RVs are completely negatively correlated, but if one **increases**, the other one **decreases** always instead
- $\rho > 0$ means that the two RVs are **positively correlated**, but if one increases by a lot, the other increases just by a little bit
- $\rho < 0$ means that the two RVs are **negatively correlated**, but if one increases by a lot, the other decreases just by a little bit

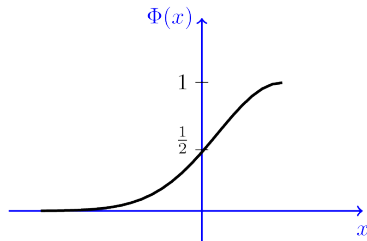
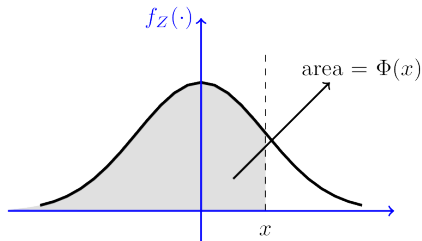


Basic Tools for today

Normal distribution

Distribution $N(\mu, \sigma^2)$ has distribution:

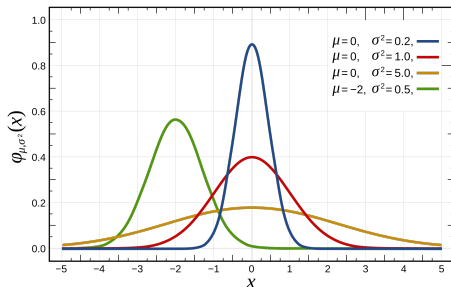
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$



Basic Tools for today

The Normal distribution has a really neat property of standardization, where any normal distribution can be turned into one with mean 0 and variance 1

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1) \quad \text{where } X \sim N(\mu, \sigma^2)$$



Theorem (Central Limit Theorem)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with mean μ and variance $\sigma^2 < \infty$. Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, as $n \rightarrow \infty$,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightarrow N(0, 1).$$

Idea: the sample mean is normally distributed, given a big enough sample size

Let's focus on data, how do we represent it?

Example

# Age	# Height	# Weight	# Bmi	▲ BmiClass
61	1.85	109.3	31.935719503287068	Obese Class 1
60	1.71	79.02	27.02369959987689	Overweight
60	1.55	74.7	31.09261186264308	Obese Class 1
60	1.46	35.9	16.841808969787955	Underweight
60	1.58	97.1	38.89601025476686	Obese Class 2
59	1.71	79.32	27.126295270339593	Overweight
59	1.7	73.32	25.370242214532873	Overweight
59	1.72	85.32	28.83991346673878	Overweight



Matrix representation

$$D = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

x_{ij} : value of covariate j in observation i

Here

- n : number of observations
- p : number of covariates

Row perspective (observations):

$$\underline{x}_i^T = (x_{i1}, x_{i2}, \dots, x_{ip}) \quad \forall i \in \{1, \dots, n\}$$

Idea: x_i could be
(age:59, height: 175, weight: 88, Bmi: 30, BmiClass: overweight)



Column perspective (all values of a covariate):

$$\underline{c}_j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix} \quad \forall j \in \{1, \dots, p\}$$

Idea: c_j could be height and have these values

$$\begin{pmatrix} 174 \\ 163 \\ 189 \\ 198 \\ 188 \\ \vdots \end{pmatrix}$$



Consider the **Mean vector**

$$\underline{\bar{X}}^T = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_p)$$

In our case it could be

$$\underline{\bar{X}}^T = (46, 178, 76, 30.5, \text{overweight})$$

where the single elements are the mean of the columns, the mean of the **covariates**

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$



Think of this: any realization of a covariate (value of a variable) can be written as the sum of the mean plus whatever is left

$$\underline{c}_1 = \bar{x}_1 \underline{1} + \underline{d}_1$$

where

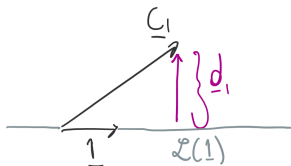
$$\underline{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$



Something neat about $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$ is that it can be written as

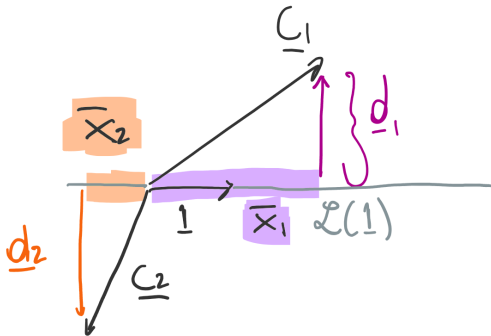
$$\Pi_{c_1 | \mathcal{L}(\underline{1})} = \frac{\langle \underline{1}, \underline{c_1} \rangle}{\|\underline{1}\| \|\underline{1}\|} \underline{1}$$

which is the projection of the vector on the linear space where **all elements are the same**



Datasets

Here is an example of two covariates projected on the linear space of the mean! The linear space is the same for both, what changes is magnitude and direction of the projection



Now we can continue this by writing out $\underline{d}_j$ as

$$\underline{d}_1 = \underline{c}_1 - \bar{x}_1 \underline{1} = \begin{pmatrix} x_{11} - \bar{x}_1 \\ x_{21} - \bar{x}_1 \\ \vdots \\ x_{n1} - \bar{x}_1 \end{pmatrix}$$

In our example, $\underline{d}$ for the covariate height would be $\begin{pmatrix} 174 - 178 \\ 163 - 178 \\ 189 - 178 \\ 198 - 178 \\ 188 - 178 \\ \vdots \end{pmatrix}$.



we notice that the 2-norm of d closely resembles the standard error (square root of variance)

$$\|d_1\|_2 = \sqrt{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \propto \sqrt{s_{11}}$$



Main idea

any dataset can be thought of as the merge of two Linear Subspaces:

- the space of the Mean, of dimension 1, where everything is equal
- the space of Variance, of dimension $(n-1)$, orthogonal to the space of the Mean

Ever wondered why the formula of sample variance is divided by $(n-1)$ and not n ? This is why

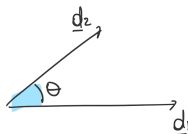
Datasets

Let's consider the interaction between two covariate vector $\underline{c}_1$ and $\underline{c}_2$

$$\underline{c}_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}, \quad \underline{c}_2 = \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix}$$

we can write $\underline{d}_1$ and $\underline{d}_2$ explicitly and represent them geometrically (like you can with any vector):

$$\underline{d}_1 = \underline{c}_1 - \bar{x}_1 \underline{1}, \quad \underline{d}_2 = \underline{c}_2 - \bar{x}_2 \underline{1}$$



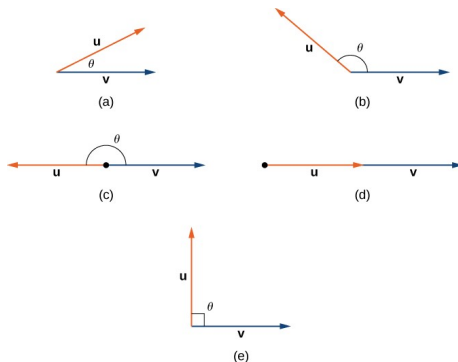
If we remember how to calculate the cosine of an angle with vectors, we find something interesting

$$\begin{aligned}\cos \theta &= \frac{\langle d_1, d_2 \rangle}{\|d_1\| \|d_2\|} = \frac{d_1^\top d_2}{\sqrt{d_1^\top d_1} \sqrt{d_2^\top d_2}} \\&= \frac{\frac{1}{n} \sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)}{\sqrt{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \sqrt{\sum_{i=1}^n (x_{i2} - \bar{x}_2)^2}} \\&= \frac{s_{12}}{s_1 s_2} = \rho_{12}\end{aligned}$$

So now we see that the correlation coefficient is nothing more than the cosine of the angle between two covariates!



Datasets



- c) covariates are the same but rescaled and opposite direction ($\rho = -1$)
- d) covariates are the same rescaled ($\rho = 1$)
- e) covariate are uncorrelated ($\rho = 0$)



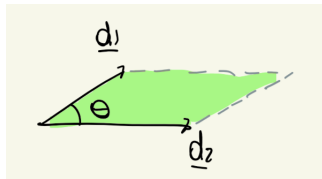
Consider the **sample covariance matrix** for a simple case with two covariates

$$\begin{aligned} S &= \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix} = \frac{1}{n-1} \begin{bmatrix} d_1' d_1 & d_1' d_2 \\ d_2' d_1 & d_2' d_2 \end{bmatrix} \\ &= \frac{1}{n-1} \begin{bmatrix} \|d_1\|^2 & \|d_1\| \|d_2\| \cos \theta_{12} \\ \|d_1\| \|d_2\| \cos \theta_{12} & \|d_2\|^2 \end{bmatrix} \end{aligned}$$

- the **generalized variance** can be represented by the determinant $\det(S)$
- the **total variance** can be represented by the trace $tr(S) = s_{11} + \dots + s_{pp}$



generalized variance



$$\begin{aligned}\det S &= \frac{1}{(n-1)^2} \left(\|\underline{d}_1\|^2 \|\underline{d}_2\|^2 - \|\underline{d}_1\|^2 \|\underline{d}_2\|^2 \cos^2 \theta \right) \\ &= \frac{1}{(n-1)^2} \|\underline{d}_1\|^2 \|\underline{d}_2\|^2 \sin^2 \theta \propto \text{Area}(\text{parallelogram}(\underline{d}_1, \underline{d}_2))\end{aligned}$$



generalized variance

- If S_{11} increases $\Rightarrow \|d_1\|^2$ increases $\Rightarrow \det S$ increases
- If $\theta_{12} = 0 \Rightarrow \det S = 0 \Rightarrow$ there is no variability between the covariates
- If $\theta_{12} = \frac{\pi}{2} \Rightarrow \det S = \max$ if I don't change S_{11} and S_{22}



total variance Imagine adding a covariate to the mix

- the total variance will increase because the trace will get a new element $tr(S) = s_{11} + \dots + s_{pp} + s_{p+1,p+1}$

You might be thinking: "Mh, we haven't talked about probability at all in this"

And you'd be correct

That's because I lied to you



90% of Statistics is Linear Algebra, 10% is Probability

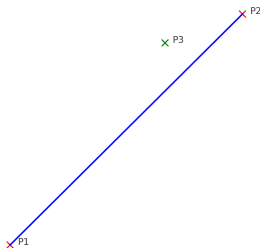




And that's why we need to talk about Linear Algebra just a little more

Least Squares

Least Squares Problem: What do we do when we try to have too many points to fit a line?



There's no line that goes through all points, so we need to come up with something else



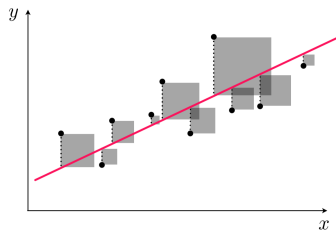
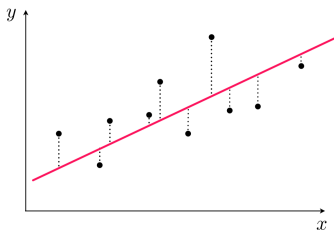
Least Squares

Let's work on a 2D plane first

Think of a point having coordinates (a, b) .

A line having the expression $b = \beta_0 + \beta_1 a$, where β_0 is the intercept and β_1 is the slope

Our objective is going to be to find the line that minimizes the sum of the distances of the points from the line



Least Squares

Let's rewrite all our points in matrix form

$$A = \begin{bmatrix} 1 & a_{11} \\ 1 & a_{21} \\ \vdots & \vdots \\ 1 & a_{m1} \end{bmatrix}, \quad \underline{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \underline{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Our minimization of the squared distances can now be expressed like this!

$$\min_{(\beta_0, \beta_1)} \|Ax - b\|_2^2 = \min_{(\beta_0, \beta_1)} \sum_{i=1}^m (a_{i1}\beta_0 + a_{i2}\beta_1 - b_i)^2$$

How do we solve this?



Least Squares

Like we would any minimization problem, we take the derivative and set it to zero

$$\nabla_{\underline{\beta}} f(\underline{\beta}) = 0$$

Compute the gradient:

$$f(\underline{\beta}) = (\underline{A}\underline{\beta} - \underline{b})^\top (\underline{A}\underline{\beta} - \underline{b})$$

$$\nabla_{\underline{\beta}} f(\underline{\beta}) = 2\underline{A}^\top (\underline{A}\underline{\beta} - \underline{b})$$

Set it to zero:

$$\underline{A}^\top (\underline{A}\underline{\beta} - \underline{b}) = 0$$



Least Squares

Let's now rewrite to get the full solution for the coefficient of our Least Squares Line

$$A^{\top}(A\underline{\beta} - \underline{b}) = 0 \quad \rightarrow \quad A^{\top}A\underline{\beta} = A^{\top}\underline{b}$$

$$\underline{\beta} = (A^{\top}A)^{-1}A^{\top}\underline{b}, \quad \text{if } A^{\top}A \text{ is invertible}$$



Why do we need this?

Linear Regression

Setup:

We assume that our variable of interest can be expressed as a linear combination of the other covariates, plus an error.

Linear Regression

Observation-wise form:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n$$

Matrix form:

$$\underline{y} = \mathbb{X} \underline{\beta} + \underline{\varepsilon}$$

where:

- $\underline{y}$: $n \times 1$ vector of **responses**
- $\mathbb{X}$: $n \times (p + 1)$ **design matrix** with first column of ones
- $\underline{\beta}$: $(p + 1) \times 1$ vector of **coefficients**
- $\underline{\varepsilon}$: $n \times 1$ **error** vector with variance $\sigma^2 I$



$$\underline{y} = \mathbb{X} \underline{\beta} + \underline{\varepsilon}$$

Remarks:

- 1 $E[\underline{\varepsilon} \mid \mathbb{X}] = \underline{0}$ (zero mean, exogeneity)
- 2 $\text{Var}(\underline{\varepsilon} \mid \mathbb{X}) = \sigma^2 I$ (homoscedasticity, no autocorrelation)
- 3 $\text{rank}(\mathbb{X}) = p + 1$ (no perfect collinearity)
- 4 usually for simplicity it's assumed that $\underline{\varepsilon} \sim \mathcal{N}(\underline{0}, \sigma^2 I)$



How do we solve this problem to find the coefficients? With Least Squares, we are still fitting a line

Solution of OLS (Ordinary Least Squares)

$$\hat{\underline{\beta}} = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \underline{y}$$



Key Idea: the reason why Statistics exists is because of $\underline{\varepsilon}$. It is a Random Variable, and it is responsible for introducing stochasticity (randomness)

Because $\underline{\varepsilon}$ is a RV, so is $\underline{y}$, and by consequence so is $\underline{\hat{\beta}}$

$$\mathbb{E}[\underline{\hat{\beta}} \mid \mathbb{X}] = \underline{\beta} \quad (\text{unbiased})$$

$$\text{Var}(\underline{\hat{\beta}} \mid \mathbb{X}) = \sigma^2 (\mathbb{X}^T \mathbb{X})^{-1}$$



It is also important to take a look at the **fitted values** (or predicted values)

$$\hat{\underline{y}} = \mathbb{X}\hat{\underline{\beta}}$$

And the difference between the real values and our predicted ones, called **residuals**

$$\hat{\underline{\epsilon}} = \underline{y} - \hat{\underline{y}}$$



Idea: if you see a **hat** on top of a variable, it means it's what we derived, what we propose, what we **think** is true, as opposed to values without a hat that represent what actually is **true**



$$\underline{\hat{\varepsilon}} = \underline{y} - \underline{\hat{y}}$$

From these we can deduce:

Estimate of σ^2 :

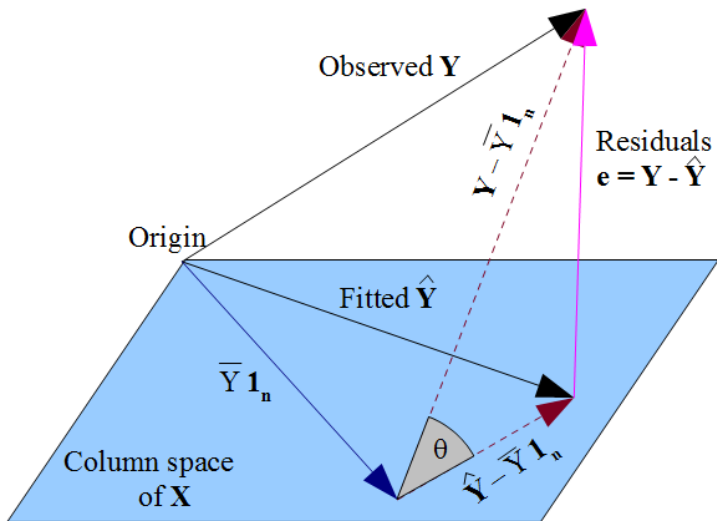
$$\hat{\sigma}^2 = \frac{\underline{\hat{\varepsilon}}^T \underline{\hat{\varepsilon}}}{n - (p + 1)}$$

Standard error of $\hat{\beta}_j$:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbb{X}^T \mathbb{X})^{-1}]_{jj}}$$



Linear Regression



let's look at an example together

Linear Regression

Dataset "mpg" on data of different models of cars

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as smf
import scipy.stats as stats

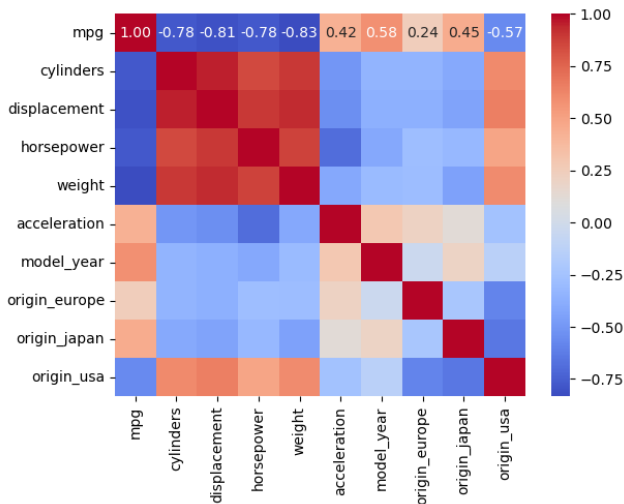
mtcars = sns.load_dataset("mpg").dropna()
mtcars.head()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino



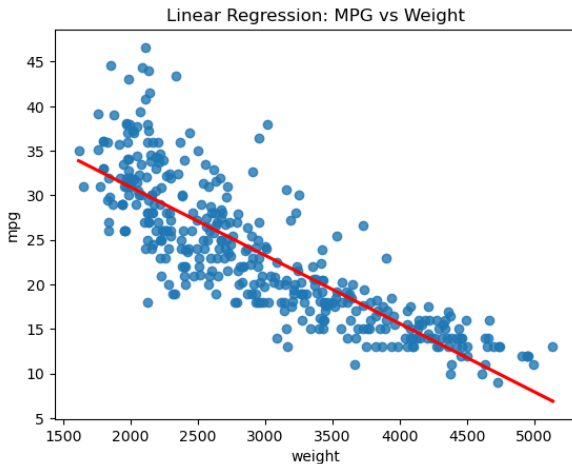
Linear Regression

first, let's look at the correlation matrix



Linear Regression

Let's immediately fit a line and see what it looks like, we are interested in the effect of weight on mpg



Linear Regression

This is what the linear regression function prints

```
model = smf.ols("mpg ~ weight", data=mtcars).fit()
print(model.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          mpg      R-squared:          0.693
Model:                  OLS      Adj. R-squared:       0.692
Method:                 Least Squares      F-statistic:      878.8
Date:                   Fri, 19 Sep 2025    Prob (F-statistic):  6.02e-102
Time:                   14:43:52           Log-Likelihood:   -1130.0
No. Observations:       392              AIC:              2264.
Df Residuals:           390              BIC:              2272.
Df Model:                1
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	46.2165	0.799	57.867	0.000	44.646	47.787
weight	-0.0076	0.000	-29.645	0.000	-0.008	-0.007

```

=====
Omnibus:                  41.682    Durbin-Watson:          0.808
Prob(Omnibus):             0.000    Jarque-Bera (JB):        60.039
Skew:                      0.727    Prob(JB):                9.18e-14
Kurtosis:                  4.251    Cond. No.                1.13e+04
=====
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.13e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Linear Regression

It's a lot, but we only care about a few things

```
model = smf.ols("mpg ~ weight", data=mtcars).fit()
print(model.summary())
```

```
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Date:                   Fri, 19 Sep 2025    Prob (F-statistic): 6.02e-102
Time:                   14:43:52    Log-Likelihood:   -1130.0
No. Observations:       392          AIC:            2264.
Df Residuals:           390          BIC:            2272.
Df Model:                1
Covariance Type:        nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	46.2165	0.799	57.867	0.000	44.646	47.787
weight	-0.0076	0.000	-29.645	0.000	-0.008	-0.007

```
=====
Omnibus:                 41.682    Durbin-Watson:      0.808
Prob(Omnibus):            0.000    Jarque-Bera (JB):    60.039
Skew:                     0.727    Prob(JB):            9.18e-14
Kurtosis:                 4.251    Cond. No.            1.13e+04
=====
```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.13e+04. This might indicate that there are strong multicollinearity or other numerical problems.



1. Significance of the coefficients

A really handy relationship that evolves from this is that you can create a test statistic that has a known distribution

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}$$

If we assume that one specific coefficient is 0, then

$$H_0 : \beta_j = 0 \rightarrow t_j \sim t_{n-(p+1)} \quad (\text{t-student distribution})$$



Confidence interval: if 0 is contained in this interval, it means that there is not enough statistical evidence to be able to confirm that β_j couldn't actually be 0

$$CI_{\alpha}(\beta_j) = \left[\hat{\beta}_j \pm t_{\alpha/2, n-(p+1)} \cdot SE(\hat{\beta}_j) \right]$$

If 0 is outside of the interval, then we can reject the hypothesis that β_j could have been 0 to begin with. In other words we are saying that we are fairly certain that β_j isn't 0, therefore significant to our model



They have an associated **p-value**, which **Idea:** we can think of as the probability of being wrong if we keep the coefficient in the model

$$p\text{-value}_j = 2 \mathbb{P}\left(T_{n-(p+1)} > |t_j|\right)$$

We generally look for p-values smaller than 5%



2. R^2 coefficient

Define the total sum of squares (total variation around the mean)

$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The regression sum of squares (explained variation)

$$\text{SSR} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2.$$

The error sum of squares (unexplained variation)

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$



Variance decomposition:

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{SSE}} + \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{SSR}}.$$

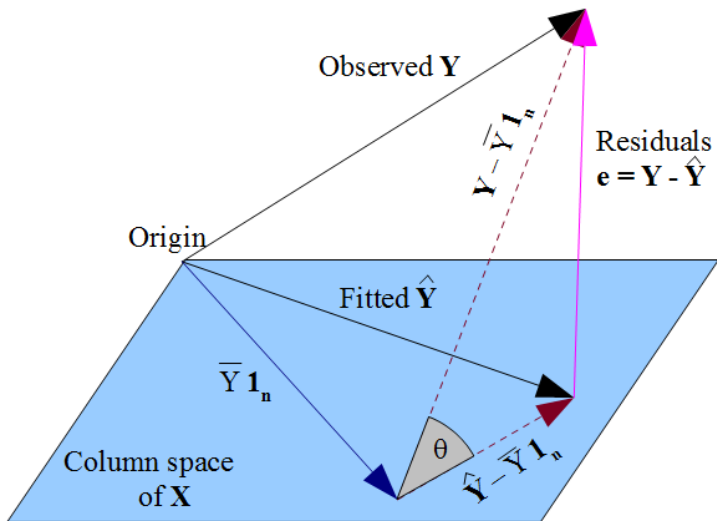
The coefficient of determination is defined as

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

Idea: R^2 is the proportion of the total variance that our model is able to capture



Linear Regression



When you do linear regression, it's important to run some diagnostics to check that what you're doing is correct and that the assumptions you made are valid

3. Normality of the model and qqplot

Idea: We take the residuals, order them, and standardize them into a distribution $N(0, 1)$. To do that, we calculate their standard counterpart that we know is distributed like a Standard Normal because of properties of the Normal distribution.

$$\hat{\varepsilon}_i = y_i - \hat{y}_i \qquad r_i = \frac{\hat{\varepsilon}_i}{\hat{\sigma} \sqrt{1 - h_{ii}}}$$

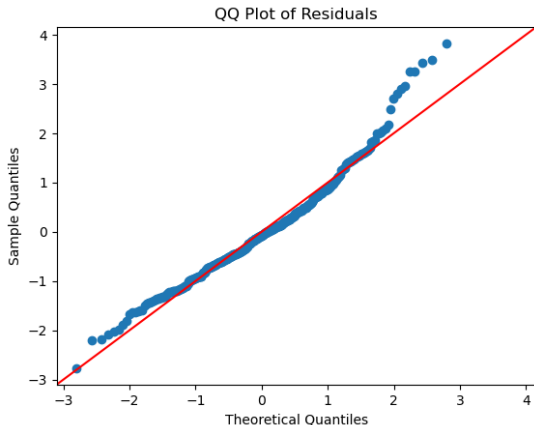
$$\hat{\sigma}^2 = \frac{1}{n - p - 1} \sum_{j=1}^n \hat{\varepsilon}_j^2$$

$$H = X(X^\top X)^{-1}X^\top, \quad h_{ii} = \text{the } i\text{-th diagonal element of } H$$



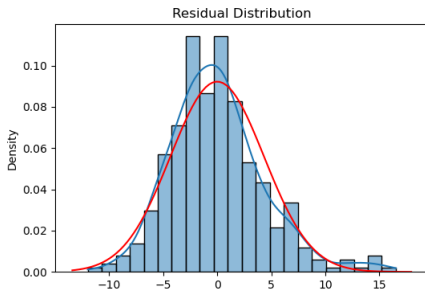
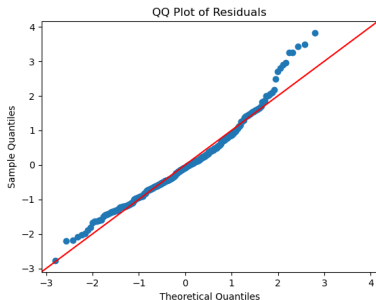
Linear Regression

This is what a **qqplot** looks like. We are comparing where our empirical quantiles, meaning the ordered values of our residuals, lie with respect to the theoretical quantiles of the standard normal



Linear Regression

To help you visualize what we are looking at, here's the distribution of the residuals.

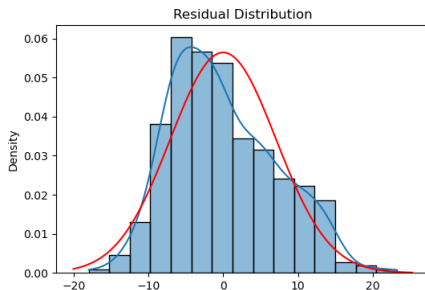
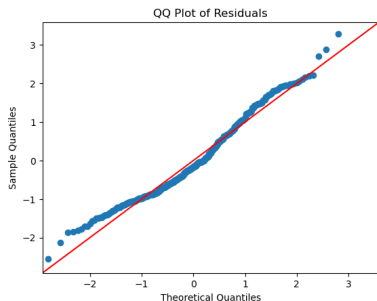


These two graphs mean the same thing

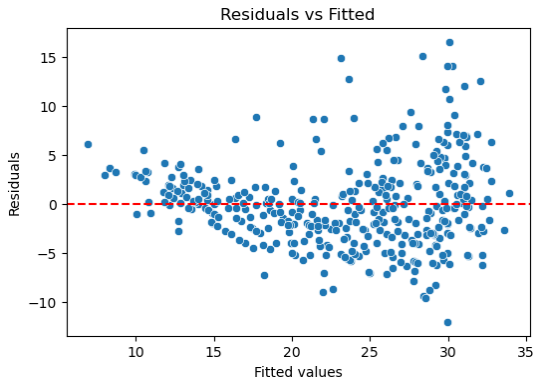


Linear Regression

For reference, this is what a **pretty bad** qqplot looks like

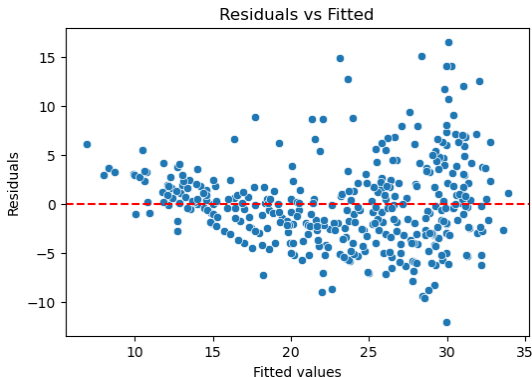


4. Residuals vs fitted Here we explore the relationship between the residuals and our fitted values, and we look out for linearity and homoschedasticity



Linear Regression

Idea: we want to see a general mean of 0 across the board, and we don't want to see cone or funnel shapes



5. Outliers We want to identify points that influence the variance so much that they skew the entire model, and possibly remove them later

To do this we look at **Cook's distance**

$$D_i = \frac{\hat{\varepsilon}_i^2}{p \hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2},$$

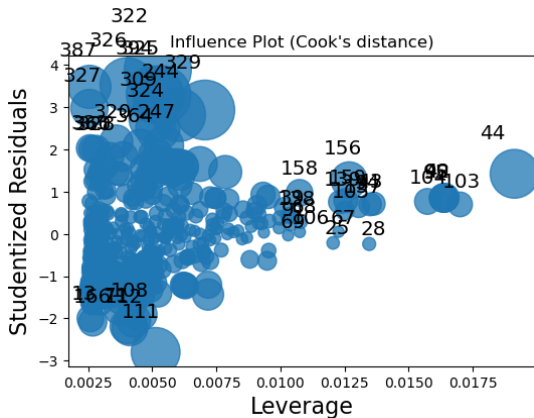
where

- $e_i = y_i - \hat{y}_i$ is the residual for observation i
- h_{ii} is the leverage (the i -th diagonal element of the hat matrix $H = X(X^\top X)^{-1}X^\top$)
- p is the number of estimated parameters



Linear Regression

Idea: Cook's distance represents how much a single point influences the model. If the value is too high (usually $D_i > 4/n$ or $D_i > 1$) then we want to remove these outliers



6. Make changes

Be flexible, and learn from what you've seen so far:

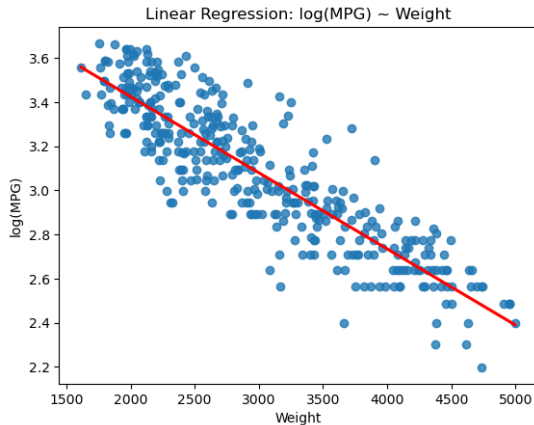
- potential non linearity observed in the data plot
- qqplot looked skewed at the end
- funnel shape in the residuals vs fitted plot
- potential outliers identified

Idea: Let's try to remove outliers and transform mpg according to a logarithmic function

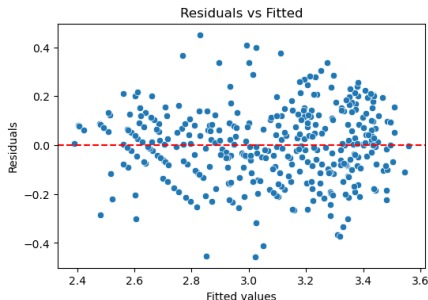
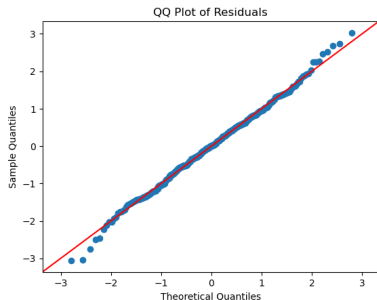


Linear Regression

Much better graph!

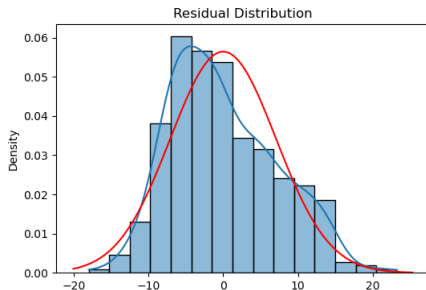
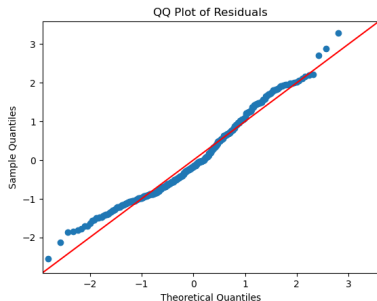


Diagnostics



Linear Regression

For reference, here's what a **really bad** qqplot plot looks like



Note: this one comes up if you fit mpg against acceleration (that we saw has low correlation with mpg in the matrix)



Linear Regression

Linear Regression models are very flexible, and can spawn a series of other models under the category of **Generalized Linear Models**

- Logistic Regression: $p(x) = \Pr(Y = 1 \mid X = x) = \frac{1}{1 + e^{-x^\top \beta}}$
- Ridge Regression: $\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \underbrace{\|y - X\beta\|_2^2}_{\text{OLS loss}} + \underbrace{\lambda \|\beta\|_2^2}_{\text{ridge penalty}}$
- Lasso Regression: $\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} \underbrace{\|y - X\beta\|_2^2}_{\text{OLS loss}} + \underbrace{\lambda \|\beta\|_1}_{\text{lasso penalty}}$

and more!



Let's now move to a more general analysis

We're not working with just linear models anymore, instead we want to assess how good our model is in general

Bias vs Variance

Let's take a look at the elements of our framework:

Consider y to be the **observed values**, $f(x)$ the model that represents the **true relationship** between our observed values and the set of **covariates** x that we chose for our model, and an **error** ε

$$y = f(x) + \varepsilon$$

Also, any **proposed model** we can think is represented by $\hat{f}(x)$, and this is because in practice we are not going to guess the **true relationship** $f(x)$



Thought Experiment: Imagine you were trying to predict the price that a house is going to be sold at

What kind of covariates could you think of collecting to create a model to predict this?

Bias vs Variance

Any covariate you thought of would in your case be added to the dataset $\mathbf{x}$, while any other you haven't considered would directly contribute to ε and its variance.

Remember: there's always going to be a covariate you haven't thought of/couldn't measure

Key Idea: no matter how many covariates we introduce, how much we tweak the model, there's always going to be an **irreducible error** ε

$$y = \hat{f}(x) + \varepsilon$$



How do we compare proposed models $\hat{f}(x)$ between each other?

Enter the **Mean Square Error**

$$\mathbb{E}\left[(y - \hat{f}(x))^2\right]$$

Idea: it's the squared difference between our proposed model $\hat{f}(x)$ and the true observed value y

Important: When we looked at linear regression, the data $\mathbb{X}$ was **fixed**, and given that we were doing linear regression.

Now instead we are taking a step backwards, the data x (as we will call it in this case) is **variable**.

Idea: you can think of it as fitting the same model to study for the same phenomenon, but with data measured on a different day, by a different person, in a different environment

Bias vs Variance

Important: it can be decomposed in the following elements

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \underbrace{\text{VAR}[\varepsilon]}_{\text{Irreducible error}}$$

Proof

$$\mathbb{E}[(y - \hat{f}(x))^2] = \mathbb{E}[(f(x) + \varepsilon - \hat{f}(x))^2] = \mathbb{E}[(f(x) - \hat{f}(x))^2] + \underbrace{\mathbb{E}[\varepsilon^2]}_0$$

Introduce $\mathbb{E}[\hat{f}(x)]$ to the difference, and we expand the terms

$$\begin{aligned}(f(x) - \hat{f}(x))^2 &= (f(x) - \mathbb{E}[\hat{f}(x)] + \mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2 \\ &= (f(x) - \mathbb{E}[\hat{f}(x)])^2 + 2(f(x) - \mathbb{E}[\hat{f}(x)])(\mathbb{E}[\hat{f}(x)] - \hat{f}(x)) + (\mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2\end{aligned}$$

The expectation of the middle element is:

$$\begin{aligned}&\mathbb{E}\left[2(f(x) - \mathbb{E}[\hat{f}(x)])(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))\right] \\ &= 2(f(x) - \mathbb{E}[\hat{f}(x)]) \underbrace{\mathbb{E}[\mathbb{E}[\hat{f}(x)] - \hat{f}(x)]}_{=0} = 0\end{aligned}$$



Bias vs Variance

Proof

because

$$\mathbb{E}[\mathbb{E}[\hat{f}(x)] - \hat{f}(x)] = \mathbb{E}[\hat{f}(x)] - \mathbb{E}[\hat{f}(x)] = 0.$$

Hence the cross term vanishes, leaving:

$$\mathbb{E}[(f(x) - \hat{f}(x))^2] = (f(x) - \mathbb{E}[\hat{f}(x)])^2 + \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2].$$

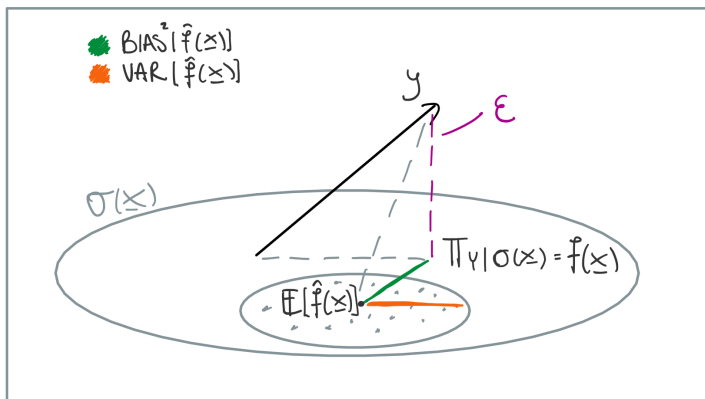
Combine

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \underbrace{\text{VAR}[\varepsilon]}_{\text{Irreducible error}}$$



Bias vs Variance

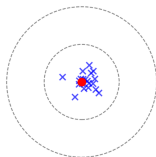
$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible noise}}$$



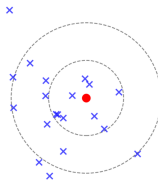
Bias vs Variance

To help you visualize further:

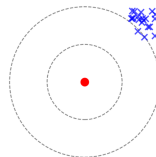
Low Bias, Low Variance



Low Bias, High Variance



High Bias, Low Variance



Idea

- **Model Variance** indicates how variable the model to change in data
- **Model Bias** indicates how close on average the model will be to the truth

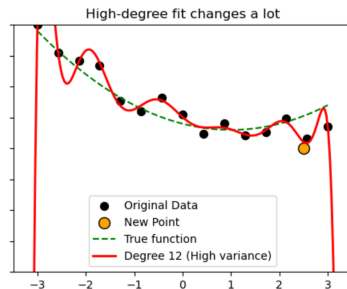
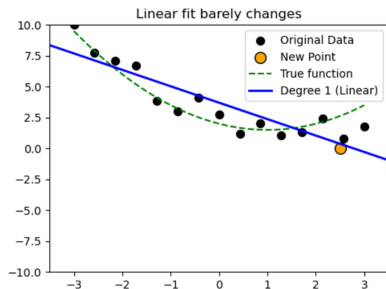
Consider an example of Polynomial Regression, meaning instead of fitting a line we want to fit an entire polynomial

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d + \varepsilon$$

if $d = 1$, then we are looking at linear regression

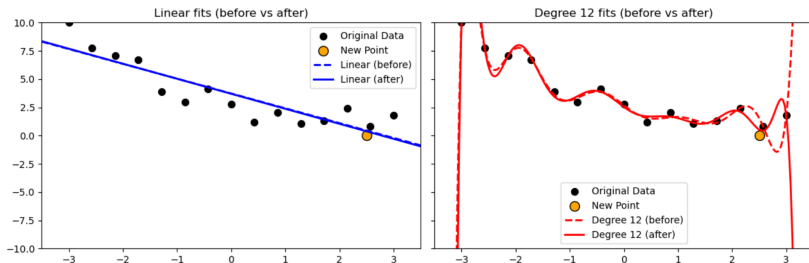
Bias vs Variance

Let's look at an example where we know the true model follows a polynomial of degree $d = 2$, and let's see what happens if we fit two regressors of $d = 1$ and $d = 12$



Bias vs Variance

Look at what happens when we introduce a **new observation** to the dataset

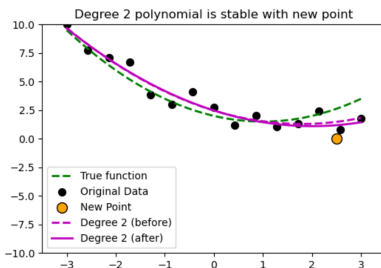
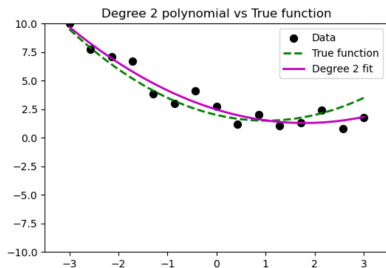


The $d = 1$ model stays the same, but it isn't super accurate still (low VAR, high Bias), while the $d = 12$ model changes significantly around the new observation (high VAR, low Bias)



Bias vs Variance

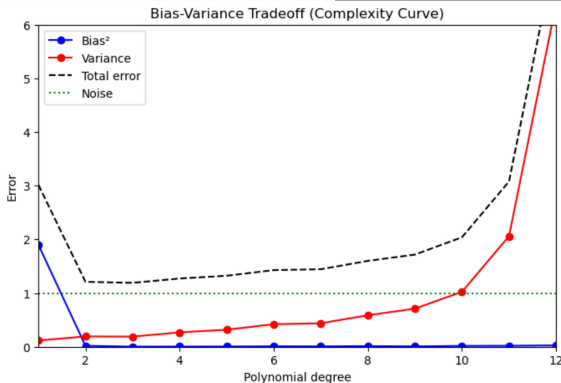
Look instead what happens if we fit a $d = 2$



It has both low VAR and low Bias

Bias vs Variance

The problems emerge when we test the model to **predict new data**

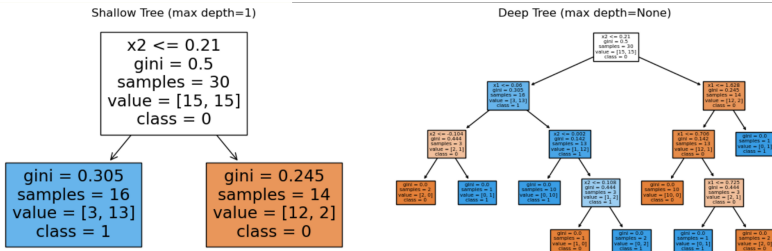


the total error of the model on new data is **minimized** on the $d = 2$ model

This type of analysis isn't limited to polynomials, it's also valid for tree models

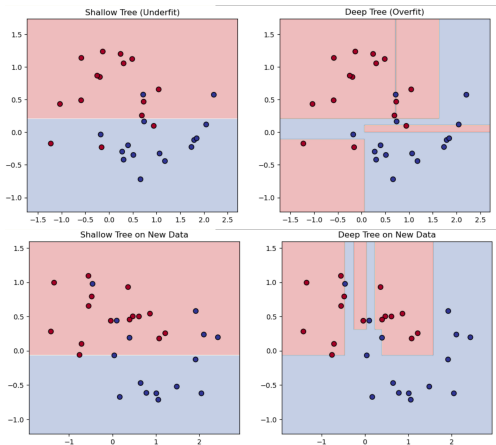
Bias vs Variance

Consider two decision trees, the left one is a very basic model makes only one binary decision, the right one instead ends up making so many splits that in the end it has one case (leaf) for every datum in the dataset



Bias vs Variance

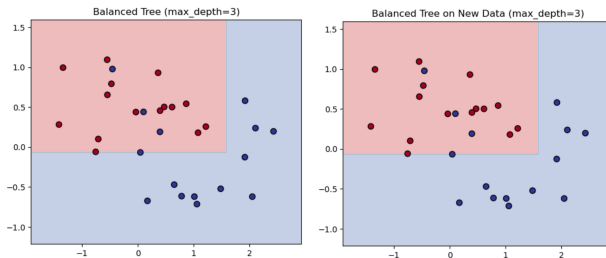
Look at what happens if we train it on two slightly different datasets from the same phenomenon



Bias vs Variance

- the **one split** tree has low variance but high bias
- the **n leaves** tree has low bias but high variance

A more **balanced tree** has both low variance and bias



Like Machine Learning, Statistics takes time to master and is not an exact science, and is comprised of a myriad of tools that take time to get accustomed to. But once you do it opens the door to incredible insights about the data you are working with

Bibliography

- 052498 Applied Statistics course (Politecnico di Milano)
- TMA4268 Statistical Learning (NTNU)
- TMA4315 Generalized Linear Models (NTNU)
- <https://www.wikipedia.org/>



Feedback form!

